Module 13: How Populations Evolve - Key Points

Learning Objectives

By the end of this module, you should be able to:

1. Define microevolution using allele frequencies.
2. Compare the five mechanisms that change allele frequencies.
3. Distinguish bottleneck and founder effects.
4. Interpret directional, stabilizing, and disruptive selection.
5. Use Hardy-Weinberg as a non-evolving baseline.

Topics

- Allele frequencies as the unit of change
- Sources of variation
- Five mechanisms of microevolution
- Genetic drift and selection
- Hardy-Weinberg as a baseline

Key Terms to Know

- **Allele frequency** - Proportion of a particular allele in a population.
- **Microevolution** - Allele-frequency change across generations.
- **Mutation** - DNA sequence change that can create new alleles.
- **Gene flow** - Movement of alleles between populations.
- **Genetic drift** - Random allele-frequency change, strongest in small populations.
- **Bottleneck effect** - Drift after a sharp population reduction.
- **Founder effect** - Drift when a new population starts from few individuals.
- **Hardy-Weinberg equilibrium** - Null model for allele frequencies when no evolution occurs.

Core Contents

1. Microevolution is change in allele frequencies across generations.
2. Mutation creates new variation, while recombination reshuffles existing variation.
3. Mutation, gene flow, genetic drift, nonrandom mating, and natural selection can change populations.
4. Genetic drift is random and strongest in small populations; selection is nonrandom and tied to fitness.
5. Hardy-Weinberg equilibrium is a null model for detecting evolutionary change.

Study Tips

1. Start with the module's central claim before memorizing vocabulary.
2. Use the same-numbered lab as the evidence check for the module.
3. Explain one mechanism in words, then redraw it as a simple diagram.
4. Answer retrieval questions without notes, then revise with the terms list.

Connected Lab

- `lab-13_how-populations-evolve.md`

Generated Visuals

- Module 13: How Populations Evolve Evidence Map
- Module 13: How Populations Evolve Reasoning Sequence
- Module 13: How Populations Evolve Retrieval and Lab Check